Soft constraint: Cantera vs NN | Test IC 04

T_0 = 1317.5 K P_0 = 1.244 atm ϕ_0 = 1.984 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

1 ANN10

2 P-only DeepONet10

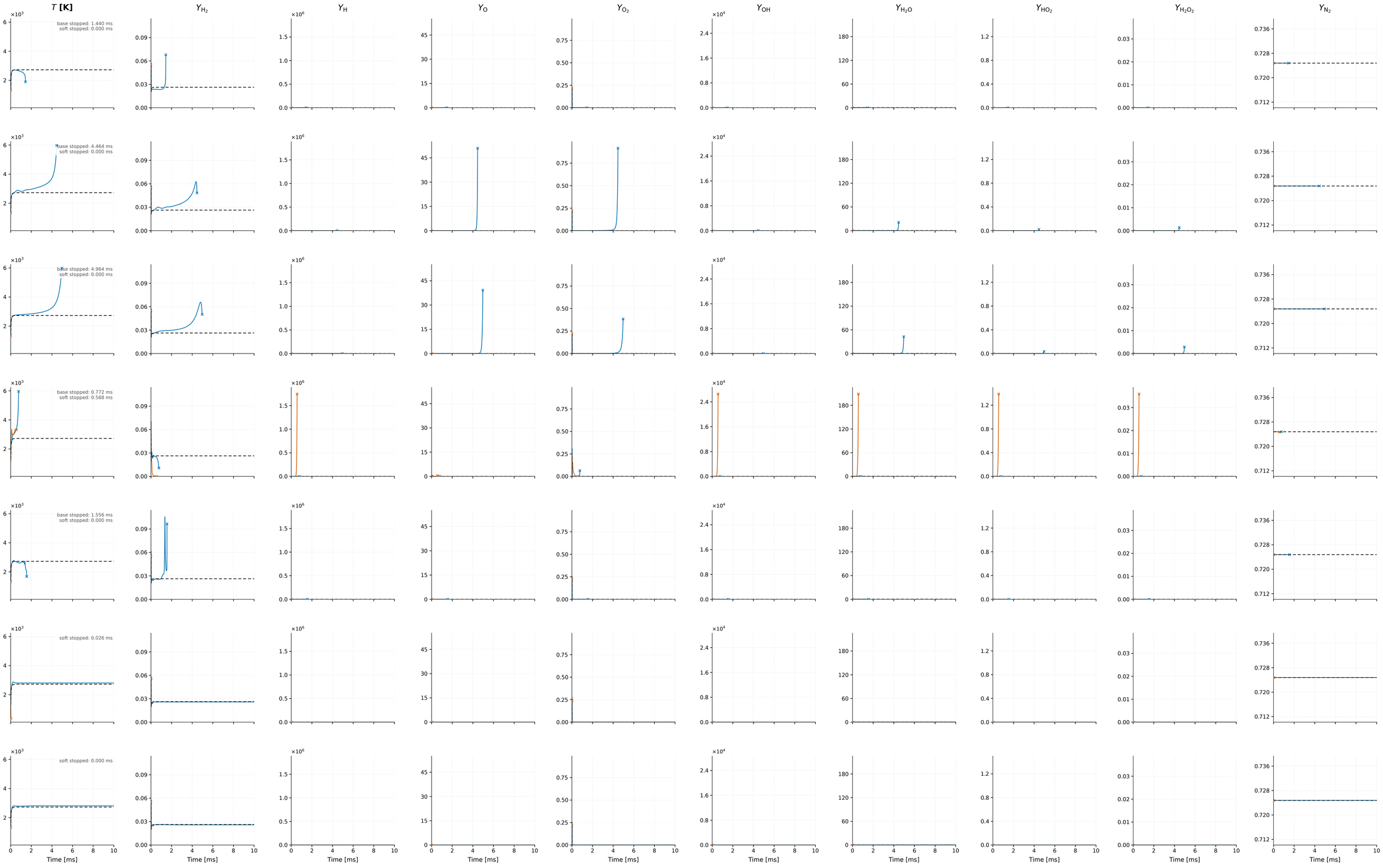
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 04 | NRMSE (util.py)

T0 = 1317.5 K P0 = 1.244 atm phi0 = 1.984 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 1.440 ms**
1 ANN10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 4.464 ms**
2 P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 4.964 ms**
3 ANN10 + P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
4 T/P DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.772 ms**
4 T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.568 ms**
5 ANN10 + T/P DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 1.556 ms**
5 ANN10 + T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
6 T/P/ϕ DeepONet10	**baseline**	0.0309	0.0188	0.215	0.559	0.164	0.406	0.0425	0.406	0.363	6.13e-16	Complete
6 T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.026 ms**
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	0.0296	0.0176	0.223	0.634	0.106	0.427	0.047	0.294	0.389	1.53e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 05

T_0 = 1049.0 K P_0 = 4.666 atm ϕ_0 = 0.674 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

1 ANN10

2 P-only DeepONet10

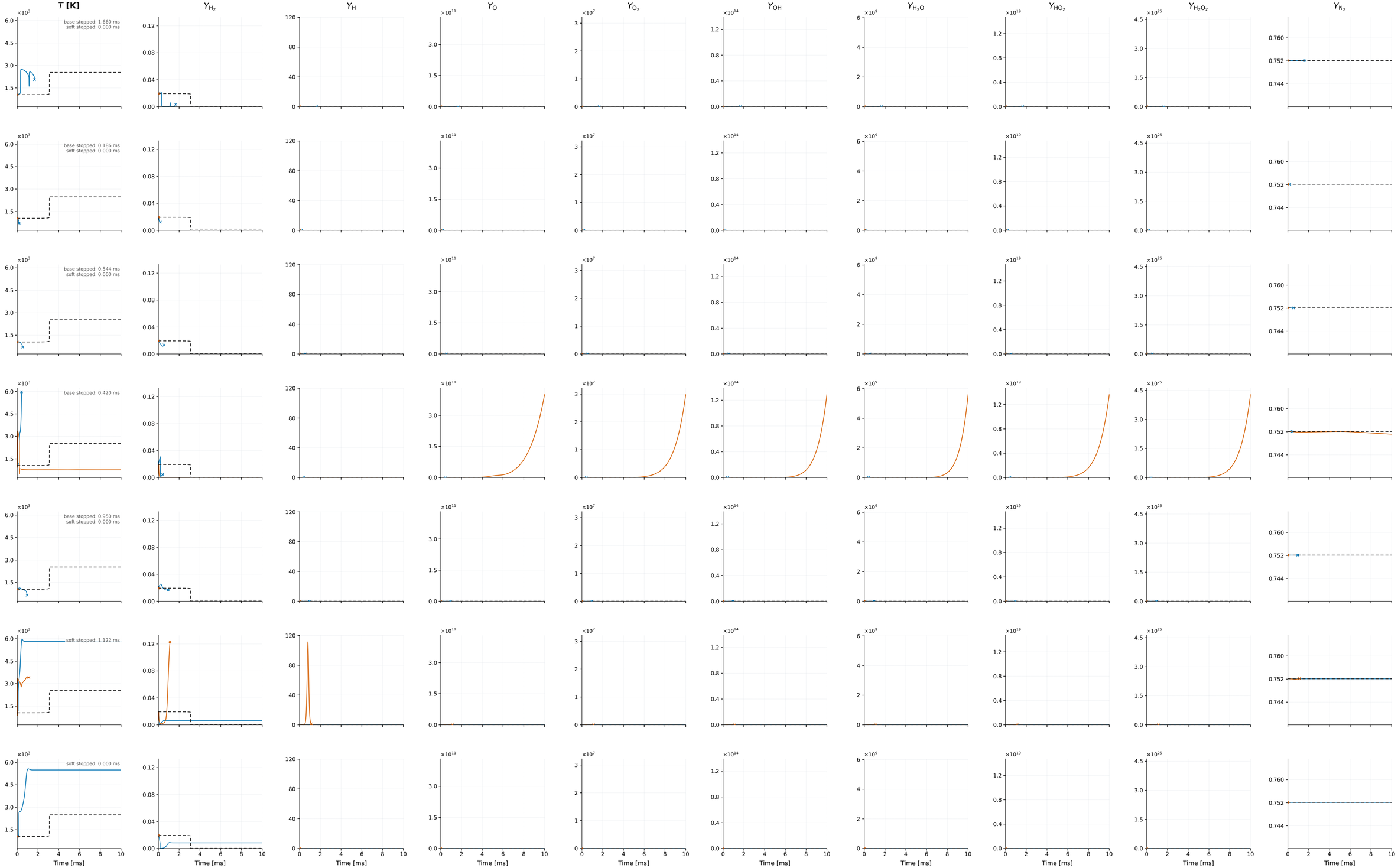
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 05 | NRMSE (util.py)

T0 = 1049.0 K P0 = 4.666 atm phi0 = 0.674 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 1.660 ms**
1 ANN10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.186 ms**
2 P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.544 ms**
3 ANN10 + P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
4 T/P DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.420 ms**
4 T/P DeepONet10	**soft10k**	0.669	0.986	1.48e+03	1.03e+14	4.87e+07	4.04e+15	8.00e+09	5.05e+22	5.88e+28	5.10e-04	Complete
5 ANN10 + T/P DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.950 ms**
5 ANN10 + T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
6 T/P/ϕ DeepONet10	**baseline**	1.71	0.835	29.8	1.22e+03	16.9	336	0.817	2.37	0.961	1.48e-16	Complete
6 T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 1.122 ms**
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	1.5	0.903	68.8	2.19e+03	16.8	203	0.602	2.85	0.951	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 07

T_0 = 930.6 K P_0 = 3.722 atm ϕ_0 = 0.754 | Raw self-fed rollout, $\Delta t = 2\,\mu$s

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

1 ANN10

2 P-only DeepONet10

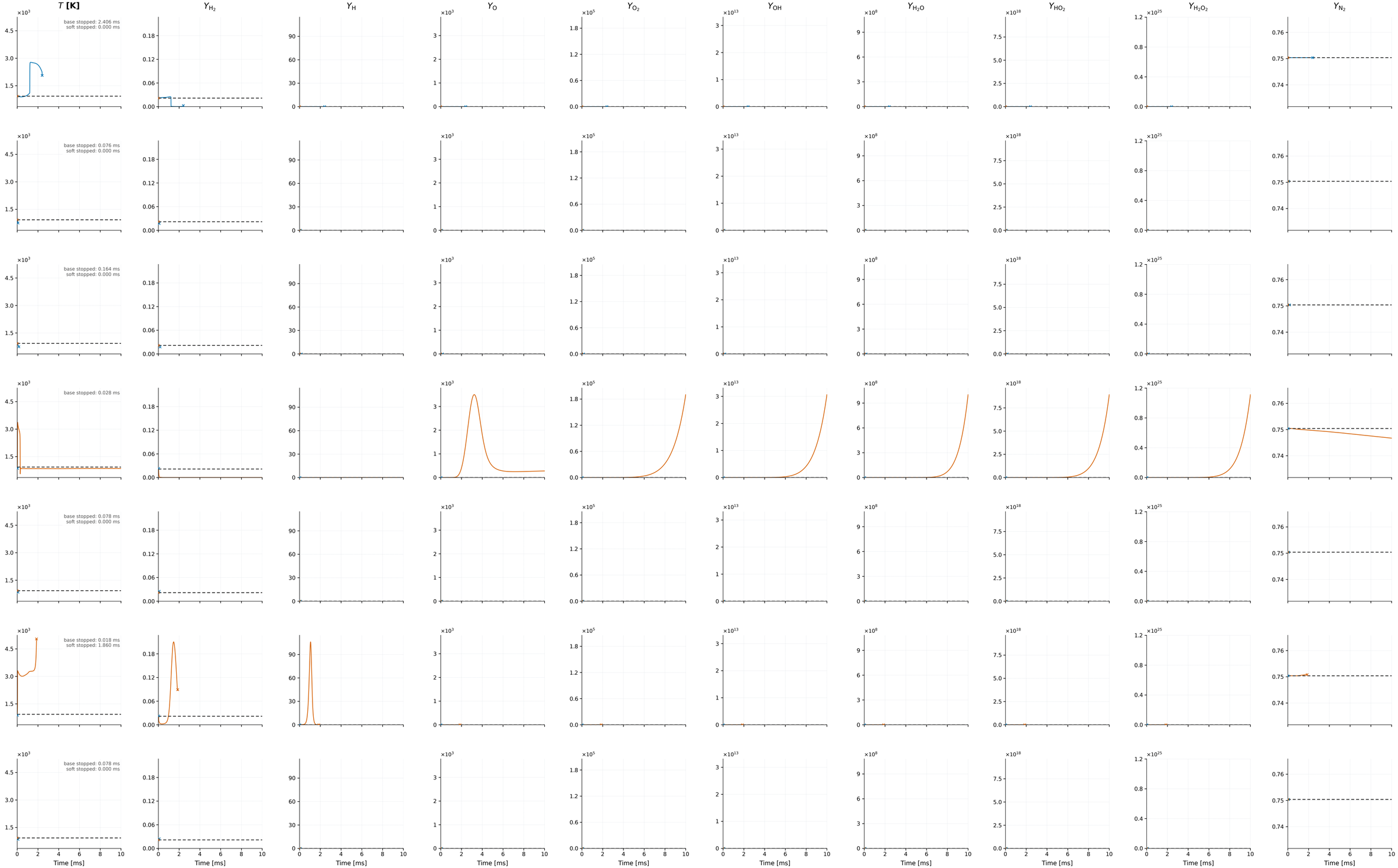
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).

All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 07 | NRMSE (util.py)

T0 = 930.6 K P0 = 3.722 atm phi0 = 0.754 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

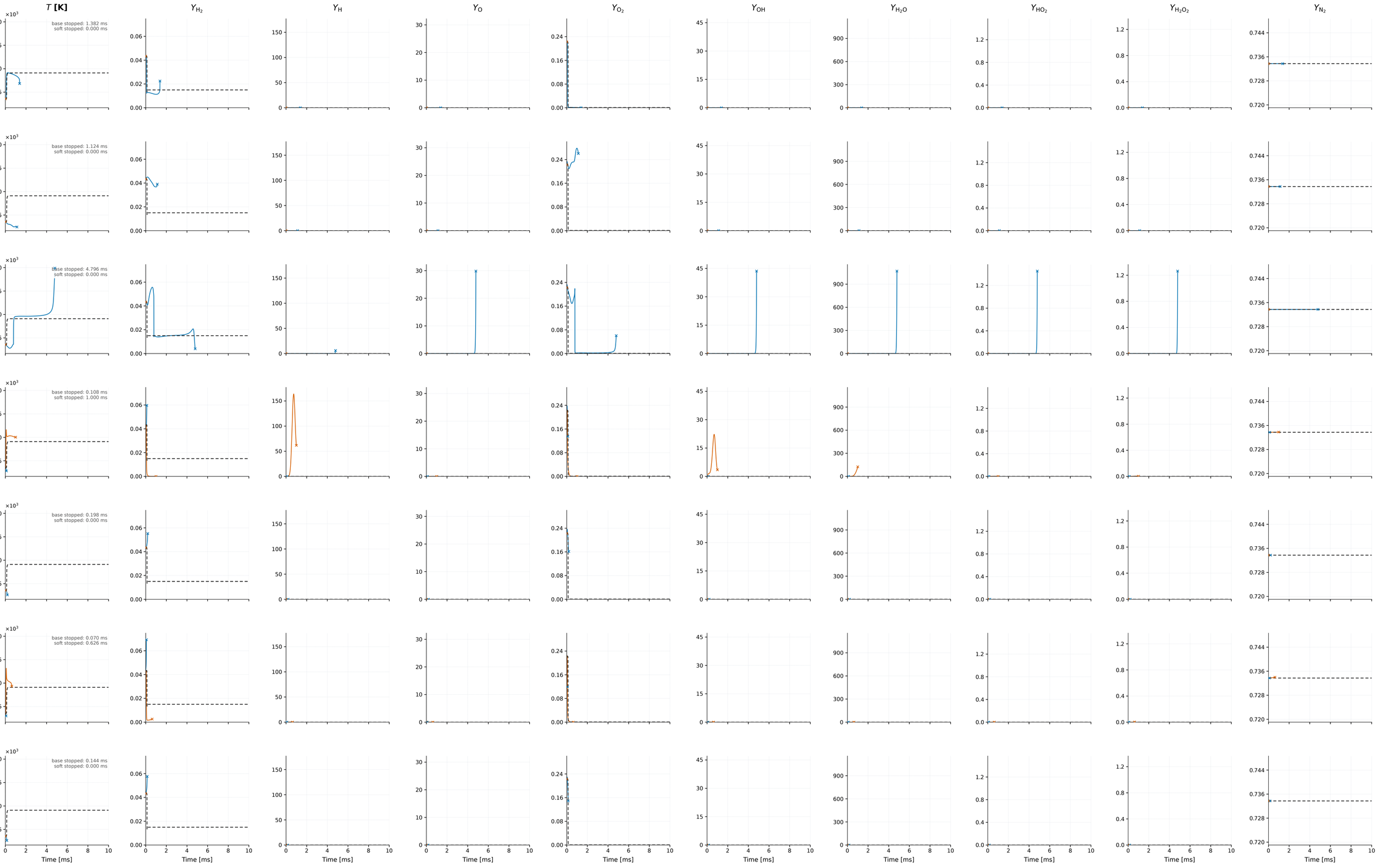
Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 2.406 ms**
1 ANN10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.076 ms**
2 P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.164 ms**
3 ANN10 + P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
4 T/P DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.028 ms**
4 T/P DeepONet10	**soft10k**	0.368	0.996	1.76e+09	1.62e+14	2.02e+05	1.57e+24	1.23e+15	4.06e+24	6.11e+30	2.67e-03	Complete
5 ANN10 + T/P DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.078 ms**
5 ANN10 + T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
6 T/P/ϕ DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.018 ms**
6 T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 1.860 ms**
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.078 ms**
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 18

T_0 = 1069.1 K P_0 = 2.170 atm ϕ_0 = 1.544 | Raw self-fed rollout, $\Delta t = 2\,\mu s$





Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 18 | NRMSE (util.py)

T0 = 1069.1 K P0 = 2.170 atm phi0 = 1.544 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 1.382 ms**
1 ANN10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 1.124 ms**
2 P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 4.796 ms**
3 ANN10 + P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
4 T/P DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.108 ms**
4 T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 1.000 ms**
5 ANN10 + T/P DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.198 ms**
5 ANN10 + T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
6 T/P/ϕ DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.070 ms**
6 T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.626 ms**
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.144 ms**
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 32

T_0 = 1140.7 K P_0 = 3.354 atm ϕ_0 = 1.463 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

1 ANN10

2 P-only DeepONet10

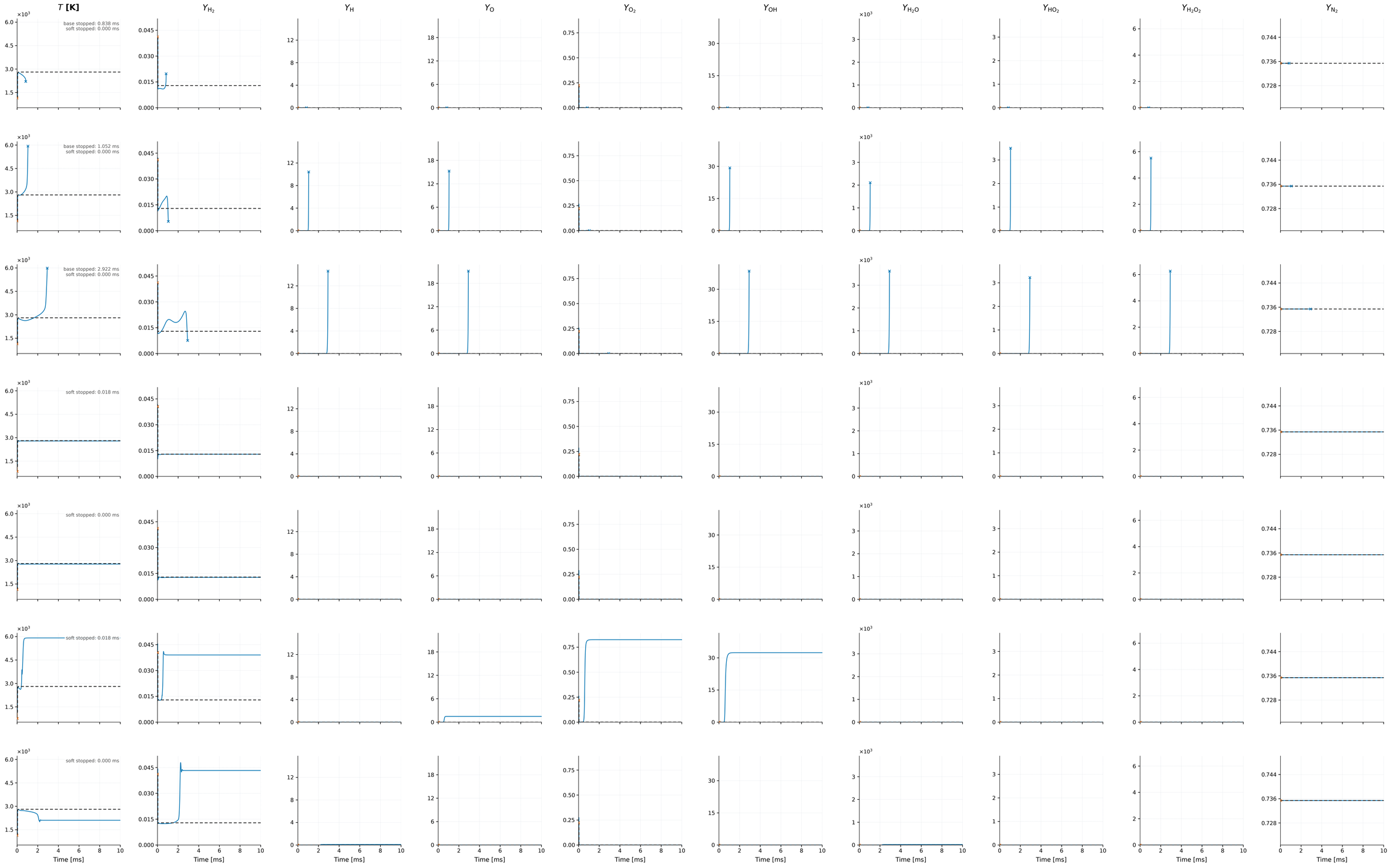
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 32 | NRMSE (util.py)

T0 = 1140.7 K P0 = 3.354 atm phi0 = 1.463 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.838 ms**
1 ANN10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 1.052 ms**
2 P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 2.922 ms**
3 ANN10 + P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
4 T/P DeepONet10	**baseline**	0.0202	0.0859	0.158	0.382	0.588	0.0956	0.0328	1.22	0.997	1.06e-15	Complete
4 T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.018 ms**
5 ANN10 + T/P DeepONet10	**baseline**	0.0163	0.0579	0.204	0.378	0.395	0.172	0.0257	1.12	1.1	1.06e-15	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
6 T/P/ϕ DeepONet10	**baseline**	1.07	1.94	2.39	2.27e+03	58	3.86e+03	0.953	23.8	421	1.06e-15	Complete
6 T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.018 ms**
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	0.224	2.05	102	0.727	0.326	0.829	75.3	1.85	30.6	1.06e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 39

T_0 = 1250.1 K P_0 = 2.297 atm ϕ_0 = 0.646 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

1 ANN10

2 P-only DeepONet10

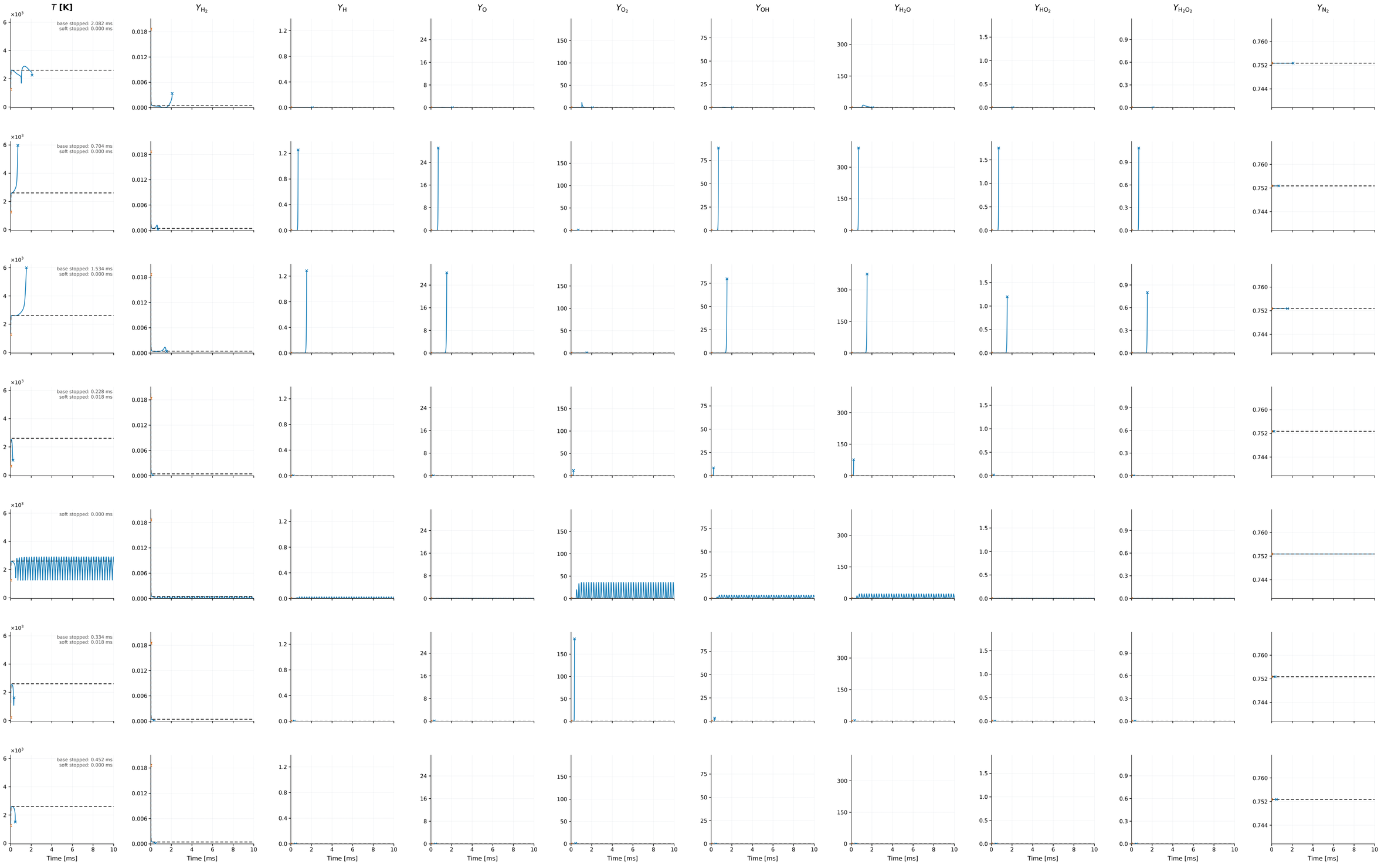
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 39 | NRMSE (util.py)

T0 = 1250.1 K P0 = 2.297 atm phi0 = 0.646 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 2.082 ms**
1 ANN10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.704 ms**
2 P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 1.534 ms**
3 ANN10 + P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
4 T/P DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.228 ms**
4 T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.018 ms**
5 ANN10 + T/P DeepONet10	**baseline**	0.225	0.427	126	12	214	167	67.7	97.8	29.2	5.90e-16	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
6 T/P/ϕ DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.334 ms**
6 T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.018 ms**
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.452 ms**
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 44

T_0 = 1031.1 K P_0 = 2.272 atm ϕ_0 = 0.770 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

	T [K]	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}
1 ANN10	base stopped: 2.236 ms soft stopped: 0.000 ms									
2 P-only DeepONet10	base stopped: 3.912 ms soft stopped: 0.000 ms									
3 ANN10 + P-only DeepONet10	base stopped: 3.514 ms soft stopped: 0.000 ms									
4 T/P DeepONet10	base stopped: 0.414 ms soft stopped: 1.314 ms									
5 ANN10 + T/P DeepONet10	base stopped: 0.870 ms soft stopped: 0.000 ms									
6 T/P/ϕ DeepONet10	base stopped: 0.374 ms soft stopped: 0.650 ms									
7 ANN10 + T/P/ϕ DeepONet10	soft stopped: 0.000 ms									

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 44 | NRMSE (util.py)

T0 = 1031.1 K P0 = 2.272 atm phi0 = 0.770 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 2.236 ms**
1 ANN10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 3.912 ms**
2 P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 3.514 ms**
3 ANN10 + P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
4 T/P DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.414 ms**
4 T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 1.314 ms**
5 ANN10 + T/P DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.870 ms**
5 ANN10 + T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
6 T/P/ϕ DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.374 ms**
6 T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.650 ms**
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	1.21	1.51	31.7	859	15.7	160	0.333	6.13	5.11	2.22e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 45

T_0 = 1181.9 K P_0 = 4.842 atm ϕ_0 = 1.099 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

1 ANN10

2 P-only DeepONet10

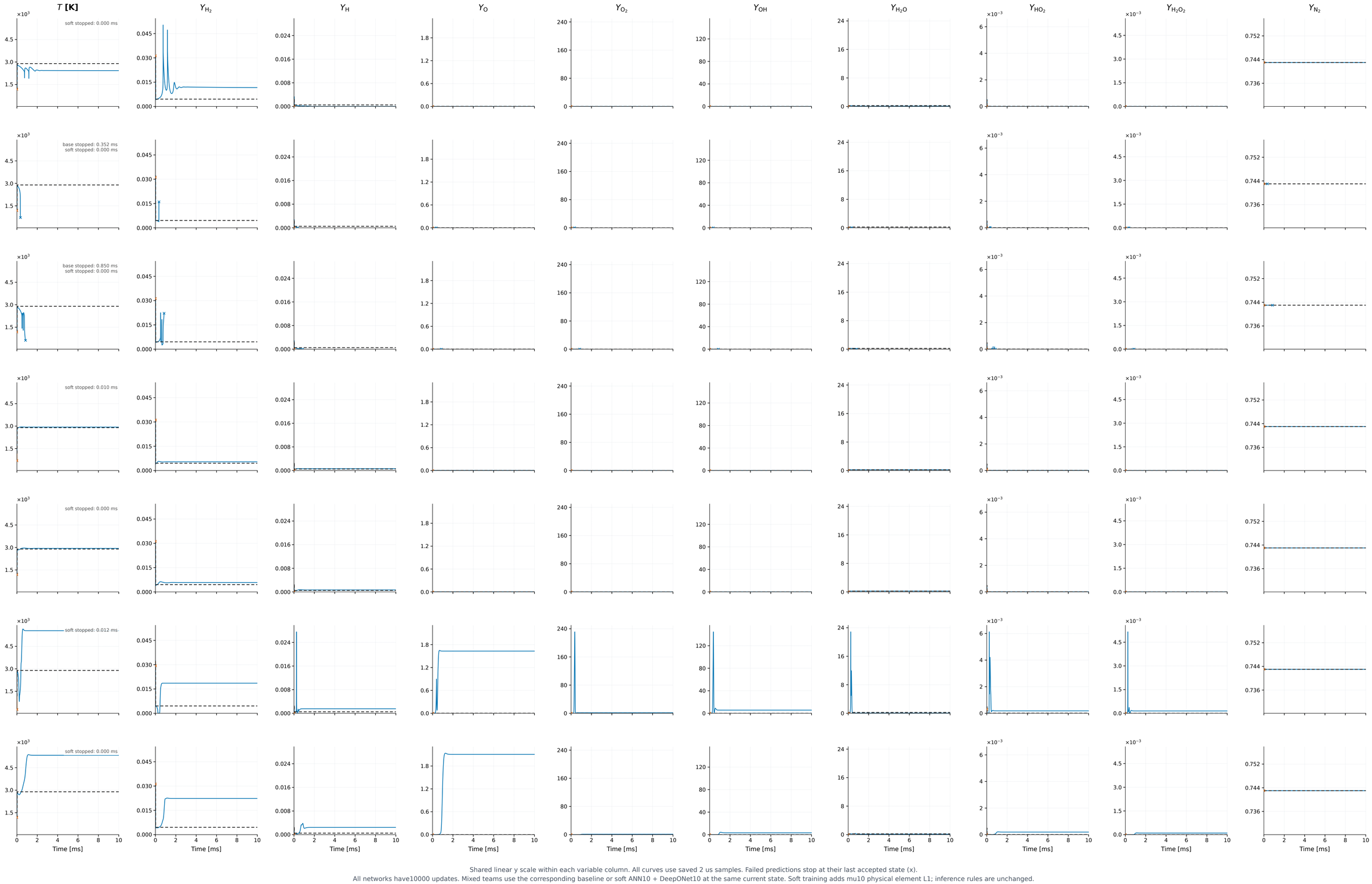
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 45 | NRMSE (util.py)

T0 = 1181.9 K P0 = 4.842 atm phi0 = 1.099 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	0.154	1.63	0.954	0.964	0.621	0.93	0.638	0.705	0.788	1.49e-16	Complete
1 ANN10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.352 ms**
2 P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.850 ms**
3 ANN10 + P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
4 T/P DeepONet10	**baseline**	0.0217	0.262	0.261	0.199	0.511	0.0576	0.0293	0.84	0.605	1.49e-16	Complete
4 T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.010 ms**
5 ANN10 + T/P DeepONet10	**baseline**	0.0214	0.315	0.312	0.16	0.505	0.116	0.0315	0.885	0.653	1.49e-16	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
6 T/P/ϕ DeepONet10	**baseline**	0.897	2.9	2.57	818	1.26e+03	788	5.08	31.5	155	1.49e-16	Complete
6 T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.012 ms**
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	0.805	3.6	3.74	1.03e+03	80.1	212	0.391	12.5	67.8	1.49e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 53

T_0 = 1257.8 K P_0 = 2.103 atm ϕ_0 = 1.799 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

1 ANN10

2 P-only DeepONet10

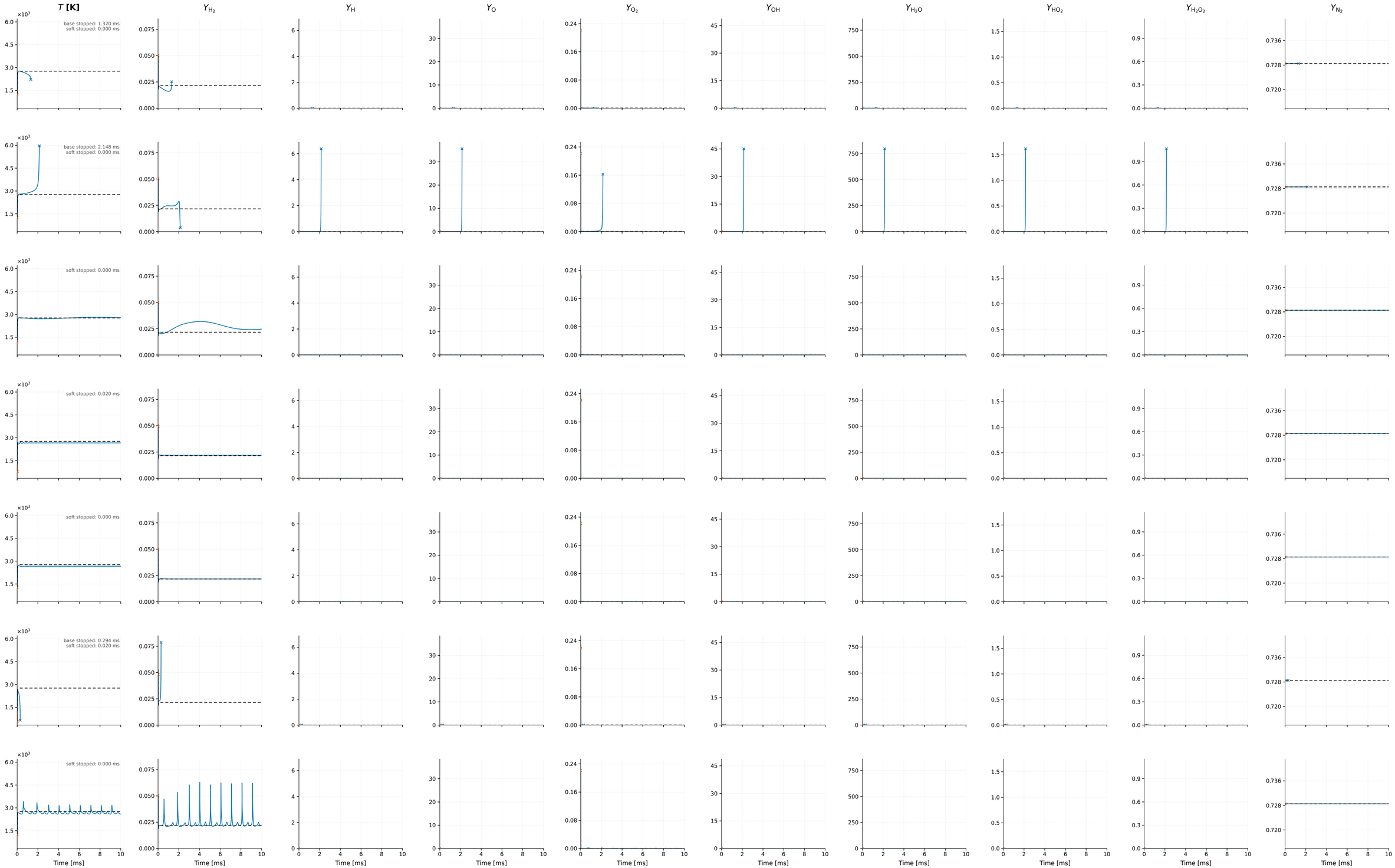
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 53 | NRMSE (util.py)

T0 = 1257.8 K P0 = 2.103 atm phi0 = 1.799 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 1.320 ms**
1 ANN10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 2.148 ms**
2 P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	**baseline**	0.0124	0.292	0.396	0.464	0.0498	0.204	0.136	0.121	0.506	7.62e-16	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
4 T/P DeepONet10	**baseline**	0.0377	0.0241	0.286	0.477	0.312	0.332	0.0163	0.592	0.456	7.62e-16	Complete
4 T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.020 ms**
5 ANN10 + T/P DeepONet10	**baseline**	0.0356	0.0124	0.425	0.598	0.201	0.426	0.106	0.415	0.373	7.62e-16	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
6 T/P/ϕ DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.294 ms**
6 T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.020 ms**
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	0.05	0.249	0.583	0.893	0.258	0.86	0.241	0.547	1.1	7.62e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 57

T_0 = 1412.0 K P_0 = 4.083 atm ϕ_0 = 0.677 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

	T [K]	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}
1 ANN10	base stopped: 2.010 ms soft stopped: 0.000 ms									
2 P-only DeepONet10	base stopped: 0.524 ms soft stopped: 0.000 ms									
3 ANN10 + P-only DeepONet10	base stopped: 0.954 ms soft stopped: 0.000 ms									
4 T/P DeepONet10	base stopped: 0.240 ms soft stopped: 0.186 ms									
5 ANN10 + T/P DeepONet10	soft stopped: 0.000 ms									
6 T/P/ϕ DeepONet10	base stopped: 0.452 ms soft stopped: 0.352 ms									
7 ANN10 + T/P/ϕ DeepONet10	soft stopped: 0.000 ms									

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 57 | NRMSE (util.py)

T0 = 1412.0 K P0 = 4.083 atm phi0 = 0.677 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 2.010 ms**
1 ANN10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.524 ms**
2 P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.954 ms**
3 ANN10 + P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
4 T/P DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.240 ms**
4 T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.186 ms**
5 ANN10 + T/P DeepONet10	**baseline**	0.332	0.813	56.4	0.932	63.3	136	59.5	57.4	32.6	1.48e-16	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
6 T/P/ϕ DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.452 ms**
6 T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.352 ms**
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	0.977	9.31	18.2	665	32.5	95.2	0.146	6.37	26.8	1.48e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 64

T_0 = 1045.3 K P_0 = 4.301 atm ϕ_0 = 0.595 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

	T [K]	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}
1 ANN10	base stopped: 1.036 ms soft stopped: 0.000 ms									
2 P-only DeepONet10	base stopped: 0.148 ms soft stopped: 0.000 ms									
3 ANN10 + P-only DeepONet10	base stopped: 0.424 ms soft stopped: 0.000 ms									
4 T/P DeepONet10	base stopped: 0.432 ms soft stopped: 0.210 ms									
5 ANN10 + T/P DeepONet10	base stopped: 0.952 ms soft stopped: 0.000 ms									
6 T/P/ϕ DeepONet10	base stopped: 0.422 ms soft stopped: 0.358 ms									
7 ANN10 + T/P/ϕ DeepONet10	soft stopped: 0.000 ms									

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 64 | NRMSE (util.py)

T0 = 1045.3 K P0 = 4.301 atm phi0 = 0.595 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 1.036 ms**
1 ANN10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.148 ms**
2 P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.424 ms**
3 ANN10 + P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
4 T/P DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.432 ms**
4 T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.210 ms**
5 ANN10 + T/P DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.952 ms**
5 ANN10 + T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
6 T/P/ϕ DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.422 ms**
6 T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.358 ms**
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	1.72	0.836	120	3.17e+03	16.6	271	0.621	3.01	0.949	1.03e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 68

T_0 = 1086.3 K P_0 = 3.702 atm ϕ_0 = 1.168 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

1 ANN10

2 P-only DeepONet10

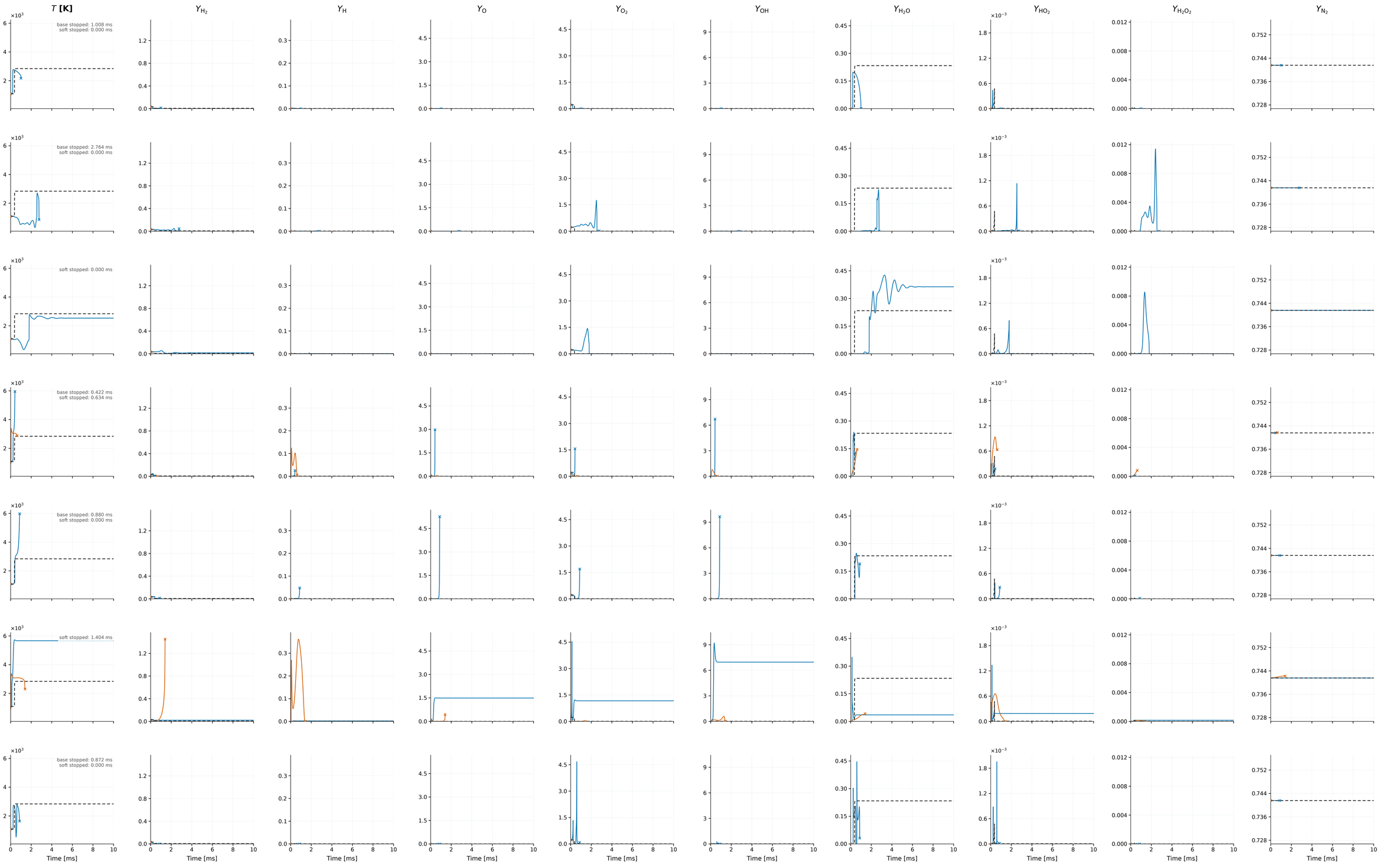
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 68 | NRMSE (util.py)

T0 = 1086.3 K P0 = 3.702 atm phi0 = 1.168 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 1.008 ms**
1 ANN10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 2.764 ms**
2 P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	**baseline**	0.295	1.67	0.514	0.978	5.87	0.866	0.624	2.66	163	4.49e-16	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
4 T/P DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.422 ms**
4 T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.634 ms**
5 ANN10 + T/P DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.880 ms**
5 ANN10 + T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
6 T/P/ϕ DeepONet10	**baseline**	1.01	1.81	2.29	1.08e+03	26.3	575	0.851	7.61	18.2	4.49e-16	Complete
6 T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 1.404 ms**
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.872 ms**
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 69

T_0 = 935.0 K P_0 = 2.045 atm ϕ_0 = 0.523 | Raw self-fed rollout, $\Delta t = 2\,\mu$s

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

	T [K]	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}
1 ANN10	base stopped: 0.128 ms soft stopped: 0.000 ms									
2 P-only DeepONet10	base stopped: 0.208 ms soft stopped: 0.000 ms									
3 ANN10 + P-only DeepONet10	base stopped: 0.280 ms soft stopped: 0.000 ms									
4 T/P DeepONet10	base stopped: 0.046 ms soft stopped: 0.010 ms									
5 ANN10 + T/P DeepONet10	base stopped: 0.188 ms soft stopped: 0.000 ms									
6 T/P/ϕ DeepONet10	base stopped: 0.320 ms soft stopped: 1.938 ms									
7 ANN10 + T/P/ϕ DeepONet10	soft stopped: 0.000 ms									

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 69 | NRMSE (util.py)

T0 = 935.0 K P0 = 2.045 atm phi0 = 0.523 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

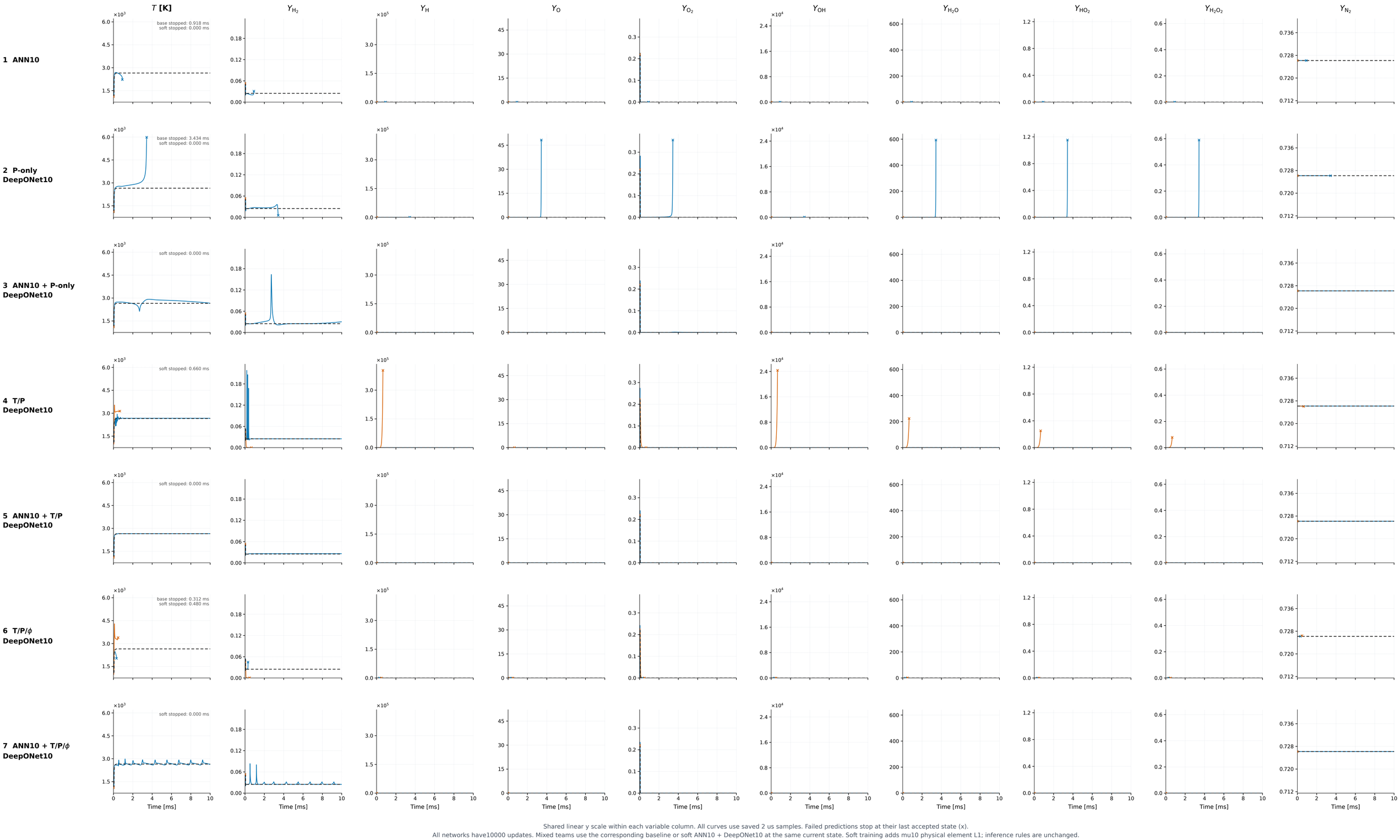
Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.128 ms**
1 ANN10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.208 ms**
2 P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.280 ms**
3 ANN10 + P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
4 T/P DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.046 ms**
4 T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.010 ms**
5 ANN10 + T/P DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.188 ms**
5 ANN10 + T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
6 T/P/ϕ DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.320 ms**
6 T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 1.938 ms**
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	5.19	0.582	9.65e+08	1.69e+11	12.3	4.23e+11	2.17e+06	829	349	1.03e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 73

T_0 = 1106.0 K P_0 = 1.911 atm ϕ_0 = 1.915 | Raw self-fed rollout, $\Delta t = 2\,\mu s$





Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 73 | NRMSE (util.py)

T0 = 1106.0 K P0 = 1.911 atm phi0 = 1.915 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.918 ms**
1 ANN10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 3.434 ms**
2 P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	**baseline**	0.06	0.5	0.694	2.01	0.0571	0.946	0.199	0.207	0.307	7.64e-16	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
4 T/P DeepONet10	**baseline**	0.0213	0.271	0.332	0.788	0.495	0.316	0.107	1.23	1.2	7.64e-16	Complete
4 T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.660 ms**
5 ANN10 + T/P DeepONet10	**baseline**	0.0125	0.0563	0.39	0.725	0.384	0.262	0.213	0.953	1.05	7.64e-16	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
6 T/P/ϕ DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.312 ms**
6 T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.480 ms**
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	0.0314	0.167	0.638	1.09	0.399	0.587	0.172	0.993	1.07	7.64e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 74

T_0 = 1222.9 K P_0 = 4.971 atm ϕ_0 = 1.766 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

	T [K]	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}
1 ANN10	base stopped: 0.666 ms soft stopped: 0.000 ms									
2 P-only DeepONet10	base stopped: 0.342 ms soft stopped: 0.000 ms									
3 ANN10 + P-only DeepONet10	base stopped: 1.354 ms soft stopped: 0.000 ms									
4 T/P DeepONet10	soft stopped: 0.010 ms									
5 ANN10 + T/P DeepONet10	soft stopped: 0.000 ms									
6 T/P/ϕ DeepONet10	base stopped: 0.436 ms soft stopped: 0.012 ms									
7 ANN10 + T/P/ϕ DeepONet10	base stopped: 0.678 ms soft stopped: 0.000 ms									

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 74 | NRMSE (util.py)

T0 = 1222.9 K P0 = 4.971 atm phi0 = 1.766 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.666 ms**
1 ANN10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.342 ms**
2 P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 1.354 ms**
3 ANN10 + P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
4 T/P DeepONet10	**baseline**	0.0146	0.0312	0.19	0.55	0.598	0.207	0.0463	1.03	0.965	1.83e-15	Complete
4 T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.010 ms**
5 ANN10 + T/P DeepONet10	**baseline**	0.0414	0.092	0.578	0.866	0.539	0.599	0.208	1.07	0.987	1.37e-15	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
6 T/P/ϕ DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.436 ms**
6 T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.012 ms**
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.678 ms**
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 76

T_0 = 1523.6 K P_0 = 3.626 atm ϕ_0 = 1.349 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

1 ANN10

2 P-only DeepONet10

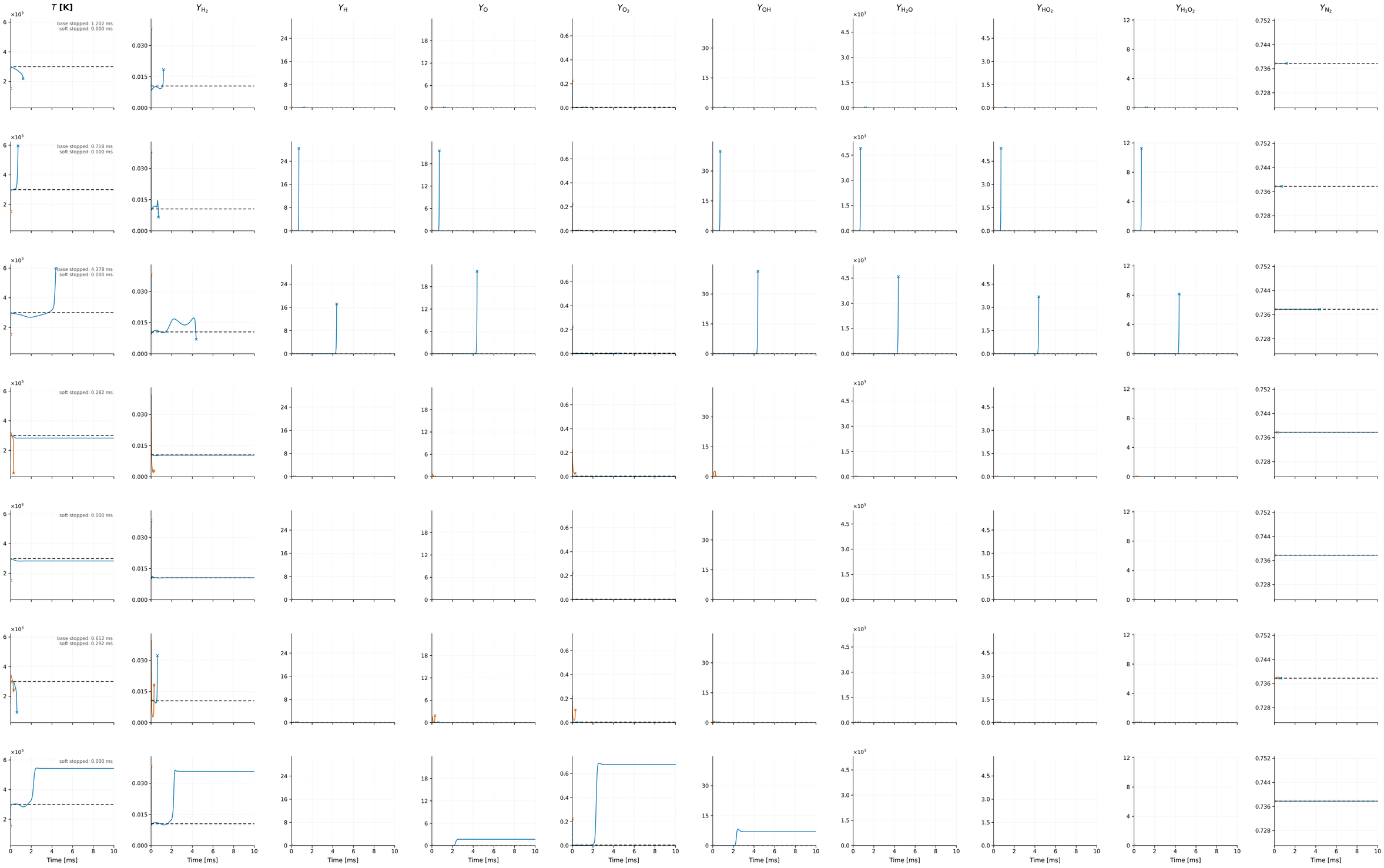
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 76 | NRMSE (util.py)

T0 = 1523.6 K P0 = 3.626 atm phi0 = 1.349 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 1.202 ms**
1 ANN10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.718 ms**
2 P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 4.378 ms**
3 ANN10 + P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
4 T/P DeepONet10	**baseline**	0.0553	0.0153	0.391	0.634	0.466	0.448	0.0778	0.653	0.534	6.02e-16	Complete
4 T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.282 ms**
5 ANN10 + T/P DeepONet10	**baseline**	0.0546	8.51e-03	0.396	0.647	0.485	0.475	0.0889	0.671	0.58	6.02e-16	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
6 T/P/ϕ DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.612 ms**
6 T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.292 ms**
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	0.714	2.11	1.21	590	89.7	351	0.664	27.1	163	6.02e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 78

T_0 = 962.3 K P_0 = 2.443 atm ϕ_0 = 1.140 | Raw self-fed rollout, $\Delta t = 2\,\mu$s

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

1 ANN10

2 P-only DeepONet10

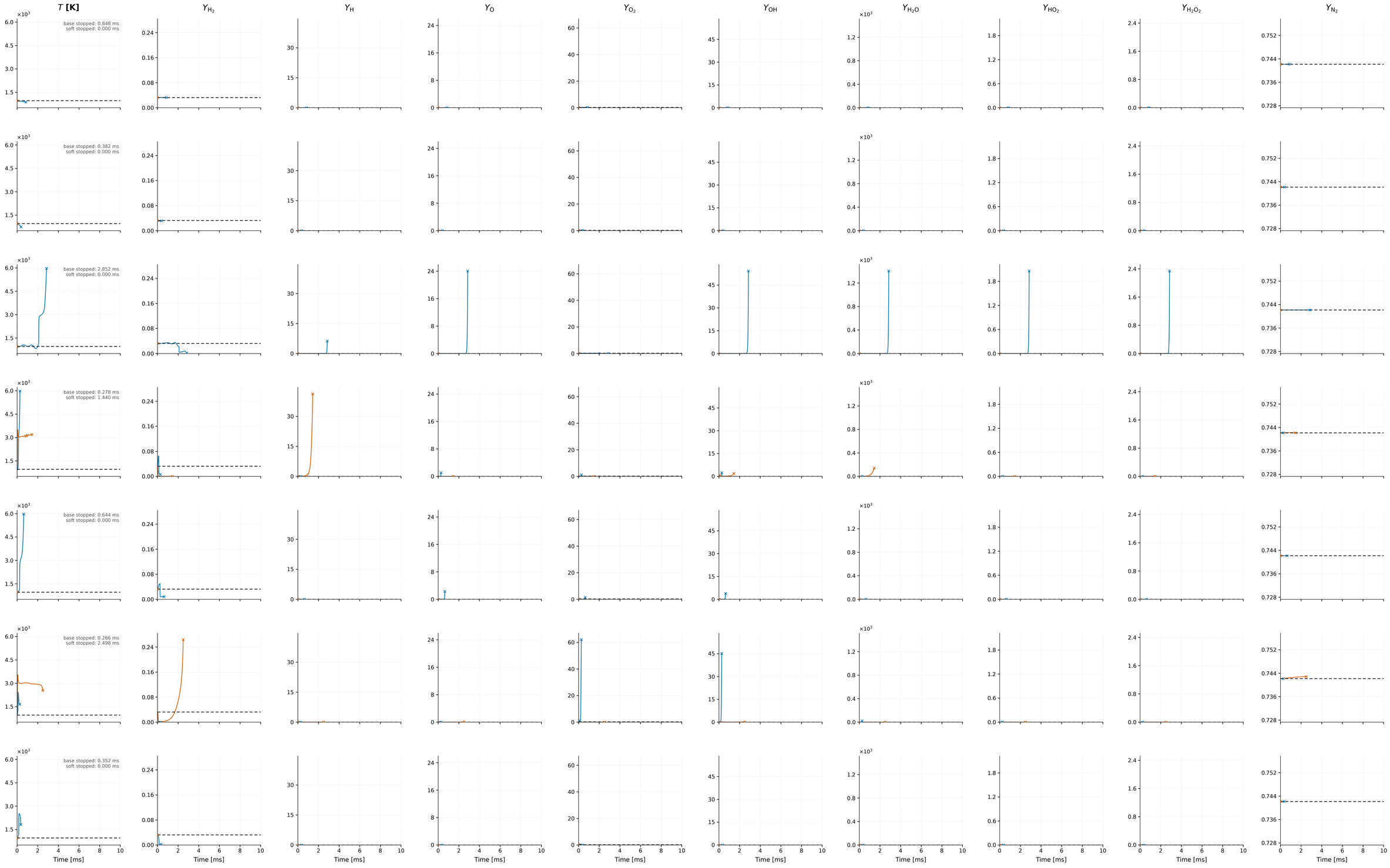
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 78 | NRMSE (util.py)

T0 = 962.3 K P0 = 2.443 atm phi0 = 1.140 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.848 ms**
1 ANN10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.382 ms**
2 P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 2.852 ms**
3 ANN10 + P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
4 T/P DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.278 ms**
4 T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 1.440 ms**
5 ANN10 + T/P DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.644 ms**
5 ANN10 + T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
6 T/P/ϕ DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.266 ms**
6 T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 2.498 ms**
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.352 ms**
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 88

T_0 = 1502.1 K P_0 = 4.980 atm ϕ_0 = 0.859 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

1 ANN10

2 P-only DeepONet10

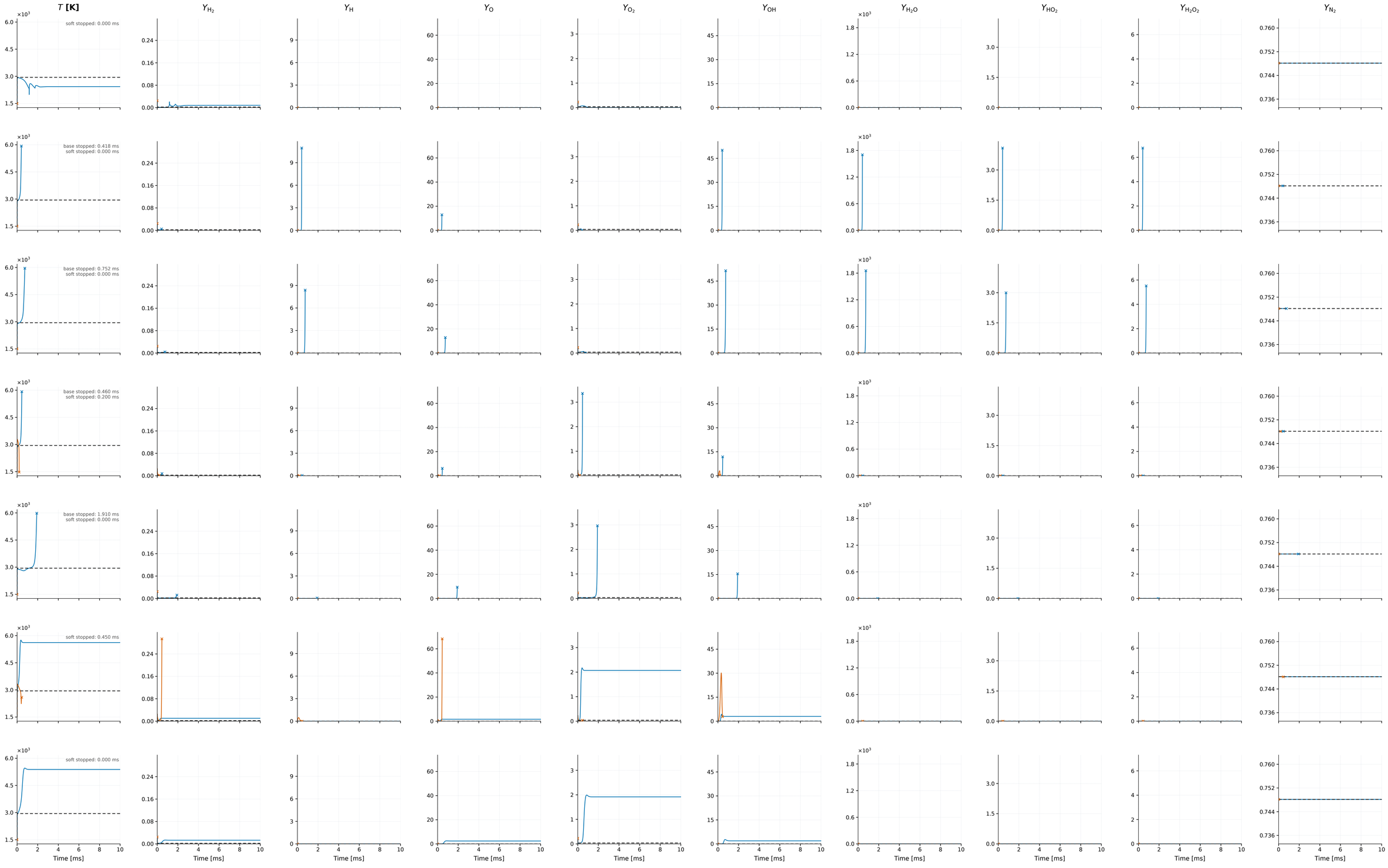
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 88 | NRMSE (util.py)

T0 = 1502.1 K P0 = 4.980 atm phi0 = 0.859 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	0.165	2.68	0.929	0.95	0.951	0.9	0.71	0.956	0.948	8.90e-16	Complete
1 ANN10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
2 P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.418 ms**
2 P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
3 ANN10 + P-only DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.752 ms**
3 ANN10 + P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
4 T/P DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.460 ms**
4 T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.200 ms**
5 ANN10 + T/P DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 1.910 ms**
5 ANN10 + T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**
6 T/P/ϕ DeepONet10	**baseline**	0.896	3.88	2.42	341	53.6	140	0.713	5.19	42.7	8.90e-16	Complete
6 T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.450 ms**
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	0.807	5.08	5.33	500	48.7	88	0.136	6.07	29.2	8.90e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.000 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.